

Smart Laboratory Catalog

Independently Developed • Full Process Control • Lifelong Service

2026

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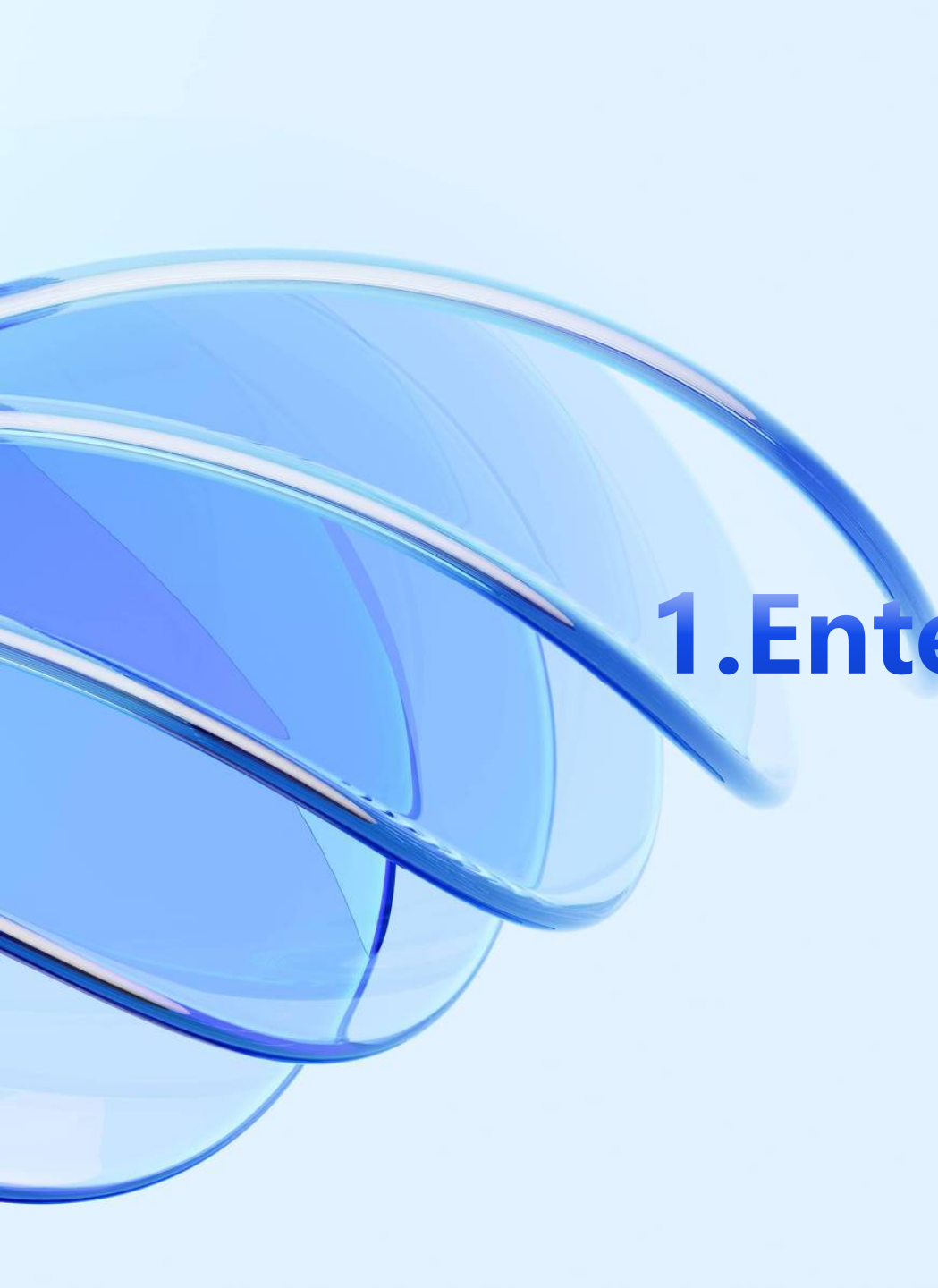


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1. Enterprise Origin

Enterprise Introduction

Guangzhou Huachuang Zhili Educational Equipment Co., Ltd. specializes in providing comprehensive and innovative integrated hardware & software solutions for educational experimental laboratories.

We deeply integrate educational technology with teaching, adopt innovative service modes to solve various difficulties in laboratory construction and experimental teaching, and promote the innovative development of education in China.

Our service network covers the whole country. Our solutions and products have been widely deployed in numerous primary, secondary and vocational colleges nationwide, winning wide recognition. Relying on premium product quality and professional after-sales system, we have gained deep trust and high praise from clients.

Our Core Teams:

Information R&D Team & Supply Chain Team

Art Design Team

Engineering Service Team

Management Team



Enterprise Introduction

Our core management team has years of experience in education industry, with abundant resources and strong capabilities in R&D innovation, lean production and market expansion.

For product R&D, we have a core expert team graduated from top universities, covering industrial structure, material science, spatial design, computer vision and algorithm development with over 10 years of industry experience. They master cutting-edge technologies and continuously drive product iteration and innovation.

Industrial Chain Advantage

We have 10 years of deep layout in Greater Bay Area industrial chain, with integrated aluminum alloy hardware supply chain to support stable long-term operation.

Enterprise Qualifications

Patents, Software Copyrights, CCC Certificates, Management System Certifications, Hardware & Software Test Reports

Corporate Mission

Lead the progress of experimental teaching and stimulate students' innovative ability. We will continuously increase R&D investment, optimize product performance and service quality, explore innovative hardware & software solutions for educational labs, support the construction of smart laboratories nationwide, and boost students' innovative thinking.





2. Scheme Presentation

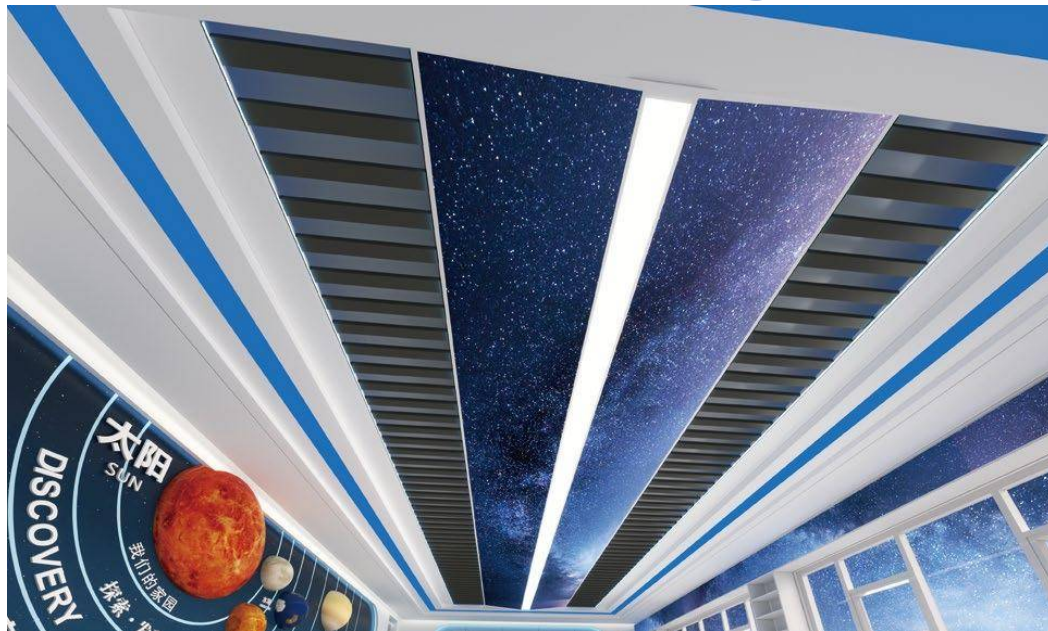
Theme Plan: Exploring the Universe



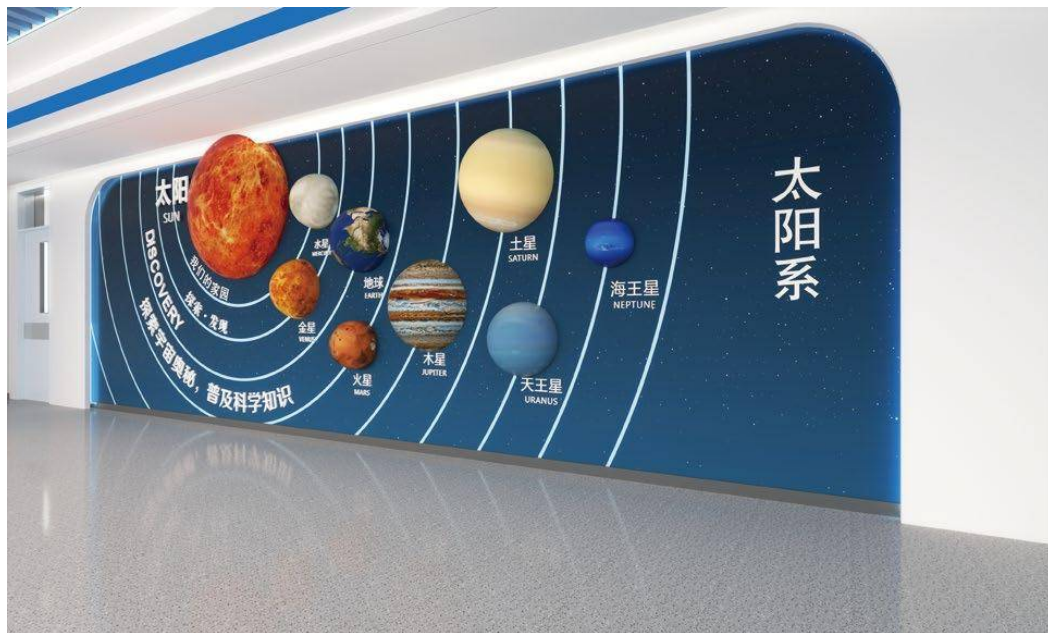
Physics Laboratory

Theme Plan: Exploring the Universe

Galaxy
Light Box
Ceiling



Solar
System
Themed
Cultural
Wall



Theme Plan: Exploration of Elements



Chemistry Laboratory

Theme Plan: Exploration of Elements

Periodic
Table of
Elements
Themed
Wall



Integrate
a variety
of design
elements



Theme Plan: Gene Decoding



Biology Laboratory

Theme Plan: Gene Decoding

Incorporate design elements of the double-helix DNA molecular structure



Specimen Themed Wall



Theme Plan: Innovative Learning Space



Theme Plan: Innovative Learning Space

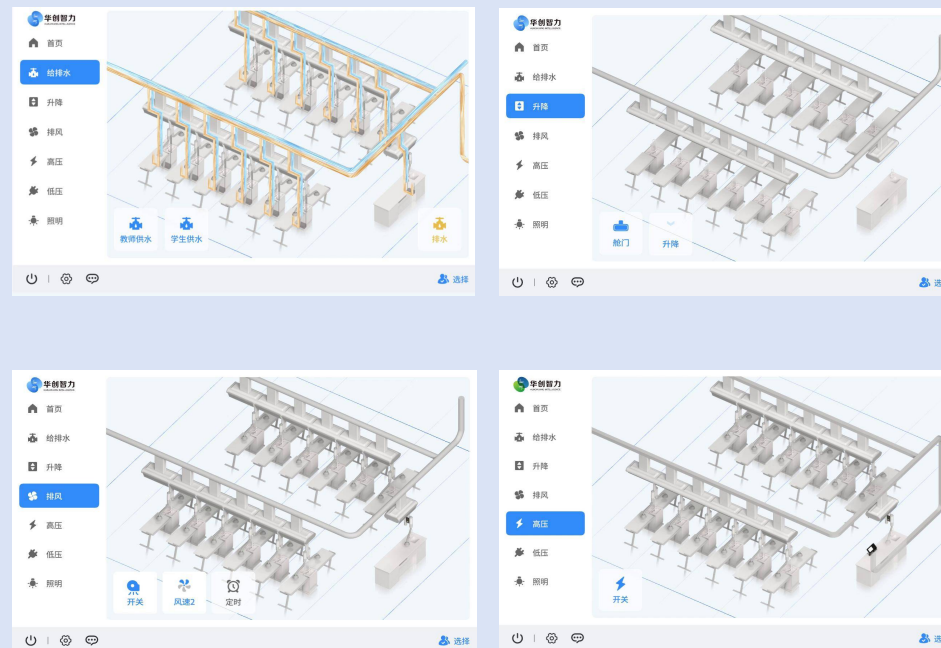




3. Product Presentation

Teacher Console

The industry-first "integrated 3D visual management and control" leverages technologies such as the Internet of Things and big data to realize one-click management of all functions including ventilation, lighting, water supply and power supply in physics, chemistry and biology laboratories.



(可视化软件画面)

Core Technology

Our desktop central control console



- Elegant and generous, intuitive and easy to learn
- Expandable IoT control function
- Compatible with smart campus systems
- Separated strong and weak current circuits for safe and user-friendly operation
- Supports data accumulation to monitor class usage status

Traditional Wall-mounted Central Controller



- Creates a sense of insecurity and discourages close operation
- Overly complicated for ordinary users to master
- Exposed wiring leads to an unattractive appearance
- Unable to collect and accumulate operational data

■ Student Power Supply

Our student power supply



- Elegant and streamlined design, intuitive and user-friendly
- Supports expandable IoT control
- Compatible with smart campus platforms
- Industry-first knob-integrated voltage display screen
- Adjust voltage via knob with real-time current display

Traditional Student Power Supply



- Unsafe
- Hard to operate
- Unattractive
- No data accumulation function

■ Uranus Hoisting Unit



The cabin door opens, and the power box swings down.

Industry-first fully concealed cabin door design

- Adopting a fully concealed cabin door design. When the cabin door is closed, it blends seamlessly with the cabin body, presenting sleek and elegant lines.
- Value: It delivers an elegant lab appearance as all equipment is completely hidden on the side of the cabin, while ensuring practicality with an unobstructed 360° viewing angle for students to observe experiments clearly. It perfectly balances functionality and aesthetic appeal.

Teacher Main Control Power Supply



The teacher main control power supply adopts a drawer-style design with a sturdy and durable sheet metal body. Equipped with safety protection, it enables unified management of power supplies for teachers and students and supports four groups of split control. It features touch keys for batch control of student terminals and dual independent DC output channels. The LCD knob allows adjustable AC and DC output with real-time parameter display. It offers wide voltage regulation range, sufficient power output and high voltage output accuracy, making it ideal for experimental teaching.

■ AI Learning Desk



Multifunctional Integration: Break the restriction of single application scenarios to achieve multi-purpose performance with one device

Core Advantages for Teaching Scenarios: Integrated with intelligent interaction, teaching system, student learning status management and other functions, it replaces the traditional combination of desks, computers, blackboards and teaching aids.

Humanized Experience: Taking both comfort and safety into consideration, suitable for high-frequency usage scenarios

Ergonomic Design: Height-adjustable desktop to fit users of all heights, rounded edge design to avoid bumps; it offers remarkably better comfort for long-hour sitting than traditional desks and chairs.

Technology Empowerment: Integrate cutting-edge technologies to build a closed-loop smart ecosystem

In-depth Integration of AI and IoT: Equipped with AI sitting posture monitoring, speech recognition, facial recognition and other modules to realize intelligent reminders, quick login, learning status analysis and other functions; it supports IoT interconnection.

Supporting Laboratory Products

Teacher Demonstration Bench

Countertop Color: Blue Green

Countertop Material: Phenolic Resin Board

Cabinet Material: Cold-rolled Steel Plate

Product Features: The whole structure is made of all steel, with all steel components treated with nano-ceramic coating for rust prevention. The countertop is crafted from high-quality solid phenolic resin board, which can effectively resist corrosion from chemical reagents. A reserved space for wiring and piping of electric circuits, water pipelines and gas circuits is designed between the cabinet and back panel of the demonstration bench, facilitating the installation of water, electricity and gas during bench assembly.



Student Lab Table

Countertop Color: Blue Green

Countertop Material: Phenolic Resin Board / Ceramic Board

Product Features: The whole unit adopts steel-aluminum structure. The countertop is made of high-quality solid phenolic resin board / ceramic board, which can effectively resist erosion by chemical reagents. The table legs and connecting profiles are manufactured from high-strength steel and processed with special crafts, delivering excellent load-bearing capacity as well as acid, alkali and corrosion resistance.



Student Lab Stool

Stool Top Color: Blue Green

Stool Surface Color: Environment-friendly Modified ABS Plastic

Product Features: The stool adopts an overall steel-plastic composite material. Its seat surface is made of ABS engineering plastic formed by integrated injection molding, which is sturdy, durable and not easy to deform. It features a height adjustment function; users can adjust the height by rotating the seat surface.



Functional Column

Functional Column Color: Blue Green

Material: PP Material

Product Features: It is integrally injection-molded with PP plastic to ensure the integrity and stability of the overall structure and deliver excellent physical properties. The access door is equipped with a special lock for convenient installation and management. Four fixing mounting points are arranged at the bottom to greatly improve equipment stability and guarantee firm and reliable operation during use.



Supporting Products for Laboratories

Universal Fume Extraction Hood



Material: Polypropylene material

Product Features: The universal fume extraction hood is specially designed for precise gas collection in laboratories. Its core joints are made of high-density PP material, enabling flexible 360° rotation, quick disassembly and assembly, as well as durability and easy cleaning. Fitted with 304 stainless steel connecting rods and embedded bearing knobs, it guarantees stable positioning. It comes with an integrally injection-molded base as standard, featuring a solid structure with no risk of base detachment. Boasting outstanding height adaptability, easy operation and a long service life, the unit meets a wide range of ventilation and exhaust requirements.

Sink Cabinet

Countertop Color: Blue Green



Material: polypropylene plastic

Product Features: The sink cabinet is integrally injection-molded from modified PP engineering plastic with acid and alkali corrosion resistance. It adopts a mortise and tenon structure reinforced with screws for stable and long-lasting performance. The front and rear are equipped with maintenance doors with rust-proof alloy locks. The PP sink is seamless with a front water retaining edge and anti-overflow holes that drain completely within 30 seconds. Pre-drilled holes are reserved for water taps and eyewash stations, and it is fitted with water-sealed stoppers to prevent backflow and clogging.

Three-way Water Tap



Material: Metal

Product Features: It balances durability and safety. Its main body is made of copper alloy coated with epoxy resin, delivering excellent rust-proof, anti-corrosion and anti-siphon performance. Adopting a scientific layout of "one high outlet and two low outlets", together with precise control via independent knobs, it can meet diverse experimental water intake and cleaning demands. High-quality ceramic cartridges are installed inside, ensuring long service life and leak-proof sealing. With standard overall dimensions, it perfectly fits all types of laboratory workbenches.

Student Power Supply



Material: Polypropylene Material

Product Features: It provides stable power output for experiments and supports remote control by teachers, greatly improving classroom management efficiency. In terms of safety, the device is built-in with dual protection components against overload and short circuit. Combined with a prominent red power indicator light, it fully safeguards the electricity safety of teachers and students as well as stable equipment operation. It features a compact design and is firmly mounted under student lab benches with screws. This concealed installation method is not only sturdy and durable, but also effectively frees up desktop space to keep the experiment countertop tidy and spacious, making it an ideal option for modern laboratories.

■ 实验室配套产品

Instrument Cabinet



Countertop Color:  Blue  Green
Countertop Material: PP Material

Product Features: This product has dimensions of 1000*500*2000mm. Its side panels and shelves are integrally injection-molded from eco-friendly modified PP material, connected via mortise and tenon joints with special plastic fasteners. Galvanized steel pipe stiffeners are embedded in the shelves. The cabinet doors and handles are injection-molded from reinforced PP, with 4mm±0.5mm tempered painted glass embedded externally; the glass on upper cabinet doors is hollowed for visibility. Door hinges are made of modified PP, and all screws are 304 stainless steel. Ventilation ports with manual adjusting devices are reserved at the top and bottom.

Medicine Cabinet



Countertop Color:  Blue  Green
Countertop Material: PP Material

Product Features: The medicine cabinet is specially designed for safe storage of laboratory reagents. The whole cabinet is integrally injection-molded from reinforced PP material, which resists acid and alkali corrosion and features a sturdy structure. It consists of upper and lower double-door compartments. The upper cabinet doors are fitted with hollow tempered coated glass, enabling clear visibility of stored chemicals and good ventilation. Two stepped PP shelves are installed inside the upper section. The lower cabinet comes with one movable shelf supported by galvanized steel pipes underneath, offering great load-bearing capacity and allowing free pull-out for flexible space layout. A ventilation port is reserved on the cabinet top for connecting external ventilation pipelines.

Hazardous Chemical Storage Cabinet



Product Features: Specially designed for secure storage of flammable liquids. The cabinet adopts a spot-welded structure with double-layer 1.0 mm steel plates and a 38 mm insulating interlayer, offering integral fireproof and explosion-proof properties. It is equipped with a three-point interlocking door lock, an electronic password lock and an optional dual-lock system for two-person custody to satisfy strict control requirements. Vent openings are arranged on both sides of the cabinet. It comes with an electrostatic grounding connector and a top pressure relief device to thoroughly eliminate static electricity and explosion risks. Both internal and external surfaces are coated with epoxy resin for corrosion resistance.

Hazardous Chemical Cabinet



Product Features: Professional Safety Storage Cabinet for Toxic Chemicals. The cabinet body is constructed with double-layer steel plates (plate thickness ≥ 1.0 mm) filled with an insulating layer of no less than 38 mm for effective fire and explosion protection. It is fitted with a three-point interlocking door lock, electronic password lock and optional dual-lock system for dual-person custody to guarantee safe access. Fire-resistant air vents are arranged on both sides of the cabinet. An electrostatic grounding connector and top pressure relief device are equipped to fully guard against static electricity and explosion hazards. Both interior and exterior are coated with epoxy resin for corrosion resistance, and the door gap is strictly controlled within 3 mm.

Summary of Advantages



Design Renderings

1. Scene Design Capability

Cosmic theme, gene theme, element theme and so on...

We provide in-depth customized thematic design with complete solution packages including high-definition renderings and animation demonstrations to integrate teaching narratives with space environments. Designs are tailored to match the campus culture of local schools, ensuring aesthetic, elegant and clearly themed solutions.



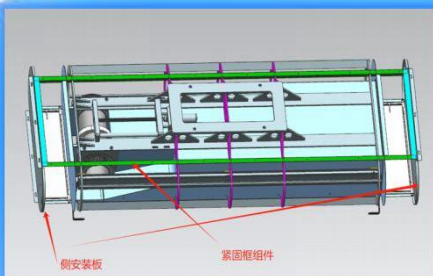
Independently developed

2. Industrial Design Capability

All products are independently designed and developed by ourselves. Every production process is under full control to guarantee product safety, allowing teachers to teach without concerns.

Value Highlights:

We have an in-house industrial design team. Our product lines adopt a systematic design language with modern and minimalist shapes that balance aesthetics and practicality.



Lifting Type

3. Structural Design Capability

Proficient in the integration of multiple processes and materials including sheet metal, profiles and injection molding, we strike an optimal balance between reliability and aesthetics.

Value: Our structural design capability serves as the engineering foundation for achieving "invisible aesthetics" and "ultimate space utilization", and acts as the core pillar of high-end customization.

Summary of Advantages



**Development
Capacity**

4. Circuit Board & Software Development Capability

Full independent R&D of integrated hardware and software with a strong research and development team.

Value: We develop self-owned PCBs and embedded programs for hardware. For software, we boast full-stack development capabilities covering UI to back-end systems, supporting high customization and agile iteration.



**Assembly
Capacity**

5. Production and Assembly Capacity

Processing is completed relying on the mature supply chain of the Greater Bay Area, while we focus on core assembly, commissioning and quality control. We maintain a streamlined factory scale, yet achieve an optimal balance of quality, cost and delivery speed through resource integration.

Value: Fast delivery and reliable quality control are guaranteed.



Service Capacity

6. Implementation & Service Capability

End-to-end services to ensure safe and normal operation of equipment.

Value: Build a more robust service system. Continuous process optimization and customer feedback mechanisms form the core of our long-term partnership shift from "project delivery" to "customer success".



(Scan the WeChat QR code
to view VR scene renderings)



(Scan the WeChat QR code
to view VR scene renderings)



(Scan the WeChat QR code
to view VR scene renderings)





4. Case Presentation

Case Study - Shenzhen Mingde Experimental School (Group)



Shenzhen Mingde Experimental School (Group) is an innovative public school covering a full 12-year education system from primary to junior and senior high school. Adhering to innovative educational philosophies, the school is committed to providing students with first-class modern learning environments and practical platforms.

Physics, chemistry and biology laboratories serve as core venues to cultivate students' scientific literacy, hands-on skills and innovative thinking. To meet the higher standards for safety, flexibility, efficiency and space utilization demanded by science education in the new era, improve the quality of experimental teaching, and cater to diverse inquiry-based experiment courses, the school has introduced laboratory lifting systems for its laboratories.

Case Study - Xingzhi School of Shenzhen Longgang Foreign Language School (Group)



Xingzhi School of Shenzhen Longgang Foreign Language School (Group) is a nine-year integrated public school with 72 teaching classes. Following Tao Xingzhi's Life Education Theory, the school has developed a featured curriculum system named "Zhile", consisting of five major curriculum clusters including morality, intellectual practice and hands-on practice. It also carries out innovative teaching activities such as "wall-free classrooms" and riverside mini-marathons.

The core of this project is the introduction of laboratory lifting systems and integrated laboratory control systems. The solutions not only address the pain points of traditional laboratories such as messy pipelines, cramped space and numerous potential safety hazards, but also take environmental psychology into consideration. They create an open, tidy, efficient and pleasant space for teaching and scientific exploration for teachers and students, fully supporting the school's innovative education and practical teaching initiatives.

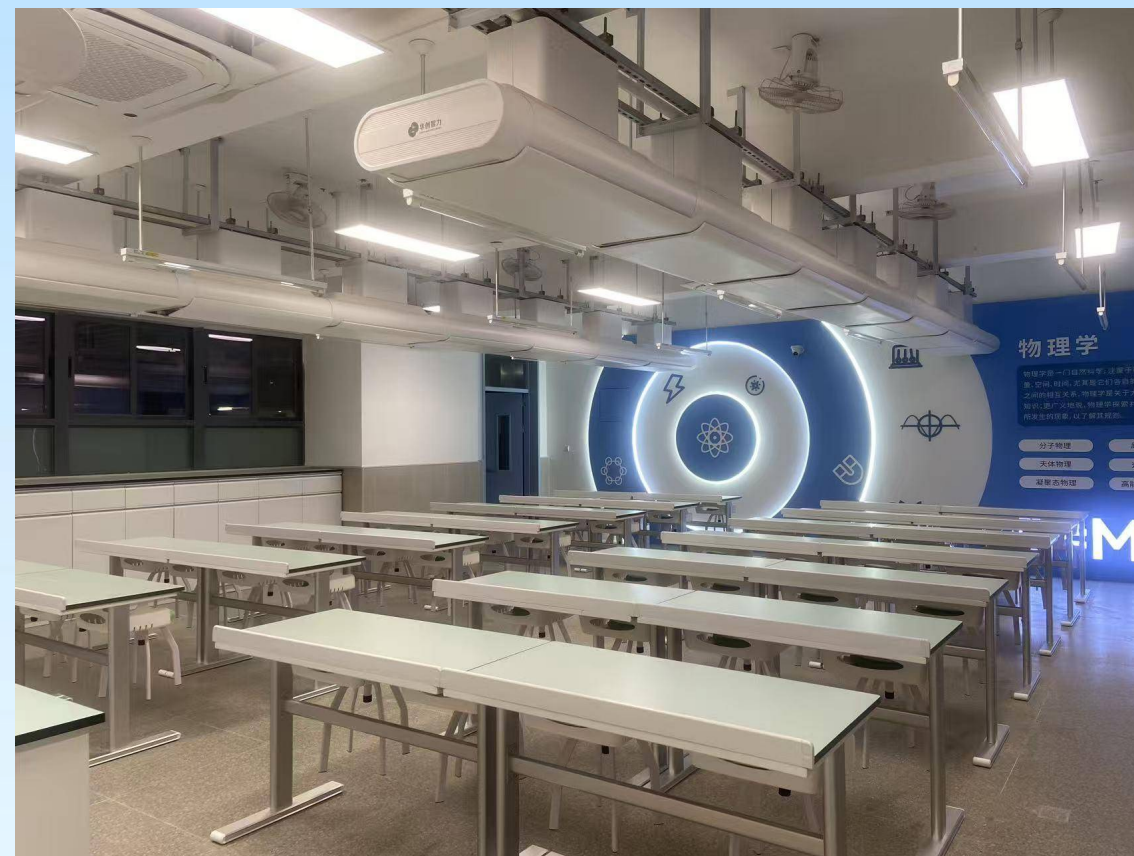
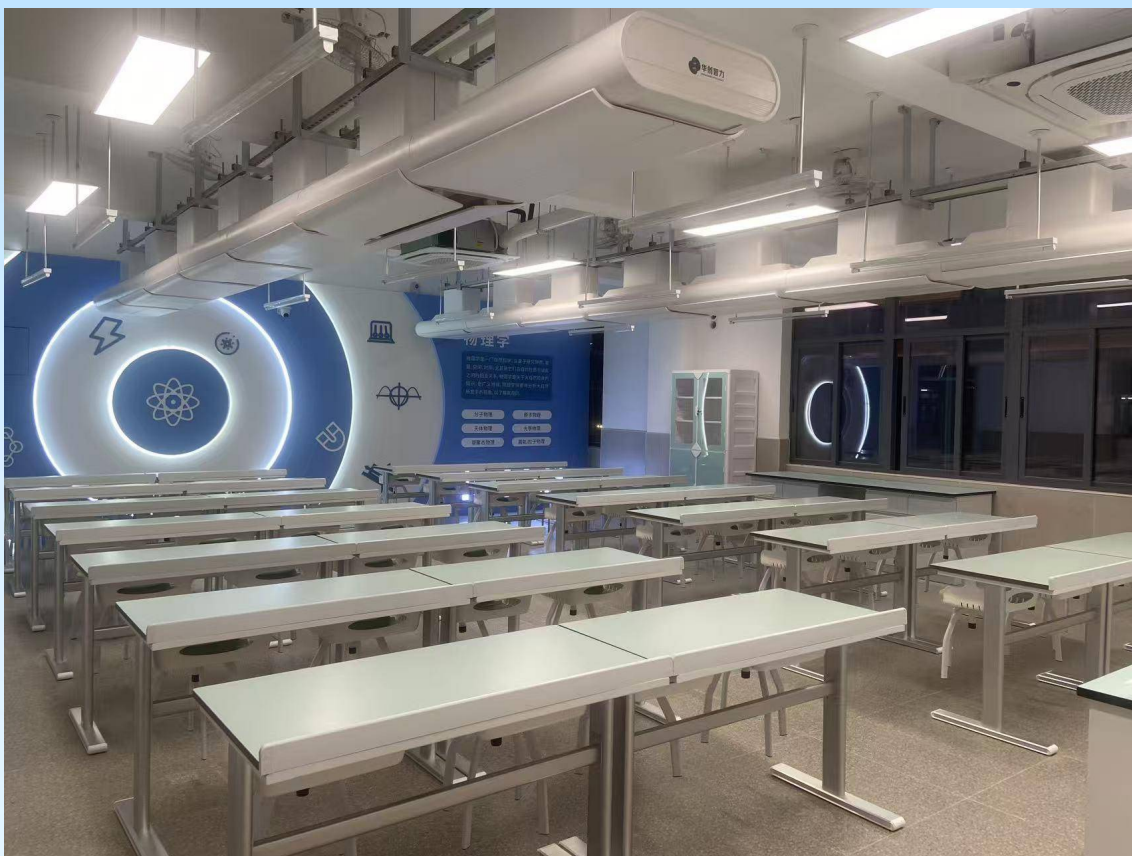
Case Study - Changyu School, Guangming District, Shenzhen



Changyu School, Guangming District, Shenzhen, is a high-standard public school established under the auspices of Guangming Senior High School (Group). It boasts abundant group school-running resources and a top-tier teaching faculty. The school prioritizes mathematics as its flagship discipline and strives to cultivate students' scientific literacy.

In addition, it fully implements the municipal requirements for the reform of science experiment teaching in junior high schools of Shenzhen. All compulsory basic experiments are guaranteed to be offered, while the number of extended experiments is increased by no less than 50%.

Case Study - Kelin School, Guangming District, Shenzhen



Shenzhen Experimental Guangming Kelin School is a high-standard nine-year integrated public school jointly built by Shenzhen Experimental Education Group and the Guangming District Government. It fully inherits the group's outstanding educational philosophy of "Sound Personality, Progressive Academic Performance" and its high-quality teaching staff. Guided by the new curriculum standards, the school implements the requirements of "disciplinary practice". Practical courses including mathematical modeling and experimental inquiry are offered across all grades to comprehensively boost students' scientific literacy.

Selected Cases

- Longhua Campus of Shenzhen Middle School
- Futian Senior High School, Shenzhen Mingde Experimental School
- Baishizhou Middle School, Nanshan District, Shenzhen
- Longgang Foreign Language School, Longhua District, Shenzhen
- Experimental School of Shenzhen Bao'an Middle School (Group)
- Tangtou School of Shenzhen Bao'an Middle School (Group)
- Junior High Department of Shenzhen Bao'an Middle School (Group)
- Taoyuanju Zhong-Australia Experimental School, Shenzhen
- Junior High Department of Shenzhen Xin'an Middle School (Group)
- Guangming Fenghuang School Affiliated to Southern University of Science and Technology, Shenzhen
- Liyuan School, Guangming District, Shenzhen
- Hongling Experimental School, Futian District, Shenzhen
- Yingyue Campus, Nanhai Experimental School, Nanhai District, Foshan
- Nanhai Middle School, Shenzhen Nanshan Experimental Education Group
- Jihua Experimental School, Longgang District, Shenzhen
- Taiziwan School, Shenzhen Shekou Yucai Education Group
- Futian Foreign Language School (Fubao), Futian District, Shenzhen
- No.1 Experimental School of Shenzhen Xin'an Middle School (Group)
- No.2 Foreign Language School of Shenzhen Xin'an Middle School (Group)
- Xingzhi School, Shenzhen Longgang Foreign Language School
- Foreign Language School of Xi'an Jiaotong-Liverpool University Basic Education Group, Futian District, Shenzhen
- Haiwang School, Bao'an District, Shenzhen
- Longtian School, Bao'an District, Shenzhen

Thanks!